1 I

1.1 (1)

Using the formula for distributing M identical balls into N numbered boxes, we obtain

$$W(M) = \binom{N+M-1}{M} = \frac{(N+M-1)!}{M!(N-1)!}.$$

1.2 (2)

Using Stirling's approximation,

$$\begin{aligned} S &= k_B \ln W \\ &\approx k_B[(N+M)\ln(N+M) - N\ln N - M\ln M]. \end{aligned}$$

Since the total energy is

$$E = \left(M + \frac{N}{2}\right)\hbar\omega,$$

we have

$$M = \frac{E}{\hbar\omega} - \frac{N}{2}.$$

Therefore, the entropy can be expressed as a function of energy:

$$\begin{aligned} S(E) \approx k_B\Bigg[&\left(\frac{E}{\hbar\omega} + \frac{N}{2}\right)\ln\left(\frac{E}{\hbar\omega} + \frac{N}{2}\right) \\ &- \left(\frac{E}{\hbar\omega} - \frac{N}{2}\right)\ln\left(\frac{E}{\hbar\omega} - \frac{N}{2}\right) \\ &- N\ln N\Bigg]. \end{aligned}$$

1.3 (3)

Using the thermodynamic relation

$$\frac{1}{T} = \frac{\partial S}{\partial E},$$

we obtain

$$\frac{1}{T} = \frac{k_B}{\hbar\omega}\ln\left(\frac{E/\hbar\omega + N/2}{E/\hbar\omega - N/2}\right).$$

Inverting this relation gives

$$\boxed{E(T) = \frac{N\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2k_BT}\right)}.$$

2 II

2.1 (4)

In the canonical ensemble, each oscillator can occupy any of its energy eigenstates, so the quantum number n ranges from zero to infinity.

The partition function of a single oscillator is

$$\begin{aligned} Z_1(\beta) &= \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega} \\ &= e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} e^{-n\beta\hbar\omega} \\ &= \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}. \end{aligned}$$

Since the N oscillators are independent, the partition function of the whole system is

$$\boxed{Z(\beta) = Z_1(\beta)^N = e^{-N\beta\hbar\omega/2} \left(\frac{1}{1 - e^{-\beta\hbar\omega}} \right)^N}.$$

2.2 (5)

The average energy of the system is

$$E = -\frac{\partial \ln Z}{\partial \beta}.$$

Taking the logarithm of the partition function,

$$\ln Z = -\frac{N\beta\hbar\omega}{2} - N \ln(1 - e^{-\beta\hbar\omega}).$$

Therefore,

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= \frac{N\hbar\omega}{2} + \frac{N\hbar\omega}{e^{\beta\hbar\omega} - 1} \\ &= \boxed{N\hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right)}. \end{aligned}$$

2.3 (6)

The heat capacity is defined as

$$C = \frac{\partial E}{\partial T}.$$

Substituting the expression for E, we obtain

$$\boxed{C = \frac{N\hbar^2\omega^2}{k_B T^2} \frac{e^{\hbar\omega/(k_B T)}}{(e^{\hbar\omega/(k_B T)} - 1)^2}}.$$

In the low-temperature limit,

$$\lim_{T \to 0} C = 0.$$

In the high-temperature limit,

$$\lim_{T \to \infty} C = N k_B.$$

2.4 (7)

The Helmholtz free energy is

$$\begin{aligned} F &= -k_B T \ln Z \\ &= \frac{N\hbar\omega}{2} + Nk_B T \ln(1 - e^{-\beta\hbar\omega}). \end{aligned}$$

Using the thermodynamic relation

$$S = -\frac{\partial F}{\partial T},$$

the entropy is

$$\boxed{S = Nk_B \left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega} - 1} - \ln(1 - e^{-\beta\hbar\omega}) \right]}.$$

In the high-temperature limit,

$$k_B T \gg \hbar\omega,$$

we define

$$x = \frac{\hbar\omega}{k_B T} \ll 1.$$

Expanding the entropy in powers of x, we obtain

$$\frac{S}{Nk_B} = 1 - \ln x + O(x^2).$$

Neglecting higher-order terms,

$$\begin{aligned} S &\approx Nk_B \left[1 + \ln\left(\frac{k_B T}{\hbar\omega} \right) \right] \\ &= Nk_B \ln\left(\frac{ek_B T}{\hbar\omega} \right). \end{aligned}$$

Comparing this with $S \approx Nk_B \ln w(T)$, we obtain

$$\boxed{w(T) = \frac{ek_B T}{\hbar\omega}}.$$

3 III

3.1 (8)

Neglecting the zero-point energy, the thermal energy of the system can be calculated using the Bose–Einstein distribution:

$$E = \int_0^\infty g(\omega) \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \, d\omega.$$

Since $g(\omega) = A\omega^2$, we have

$$E = A\hbar \int_0^\infty \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \, d\omega.$$

Introducing the dimensionless variable

$$x = \beta\hbar\omega = \frac{\hbar\omega}{k_B T},$$

we obtain

$$E = A\frac{(k_B T)^4}{\hbar^3}\int_0^\infty \frac{x^3}{e^x - 1}\,dx.$$

Since the integral is independent of temperature,

$$E \propto T^4.$$

Therefore, the heat capacity satisfies

$$\boxed{C = \frac{\partial E}{\partial T} \propto T^3}.$$

3.2 (9)

Considering a three-dimensional system of volume $V = L^3$, we impose periodic boundary conditions:

$$L = n_i \lambda_i = n_i \frac{2\pi}{k_i},$$

where $i = x, y, z$. Thus,

$$n_i = \frac{L}{2\pi} k_i.$$

The density of states in energy space is defined as

$$g(\varepsilon) = \int \delta(\varepsilon(\vec{k}) - \varepsilon)\, dn_x\, dn_y\, dn_z.$$

Using

$$dn_x\, dn_y\, dn_z = \frac{V}{(2\pi)^3}\, d^3k,$$

we obtain

$$\begin{aligned} g(\varepsilon) &= \frac{V}{(2\pi)^3}\int \delta(\varepsilon(\vec{k}) - \varepsilon)\, d^3k \\ &= \frac{4\pi V}{(2\pi)^3}\int_0^\infty k^2 \delta(\varepsilon(k) - \varepsilon)\, dk \\ &= \frac{V}{2\pi^2}\int_0^\infty k^2 \delta(\varepsilon(k) - \varepsilon)\, dk. \end{aligned}$$

From the given dispersion relation,

$$\omega = Bk^n,$$

the excitation energy is

$$\varepsilon(k) = \hbar\omega(k) = \hbar B k^n.$$

Therefore,

$$g(\varepsilon) = \frac{V}{2\pi^2}\int_0^\infty k^2 \delta(\hbar B k^n - \varepsilon)\, dk.$$

Introducing the variable

$$x = \hbar B k^n,$$

we have

$$k = \left(\frac{x}{\hbar B}\right)^{1/n},$$

and

$$dk = \frac{1}{n}(\hbar B)^{-1/n} x^{1/n-1}\, dx.$$

Substituting these expressions into the integral, we obtain

$$\begin{aligned} g(\varepsilon) &= \frac{V}{2\pi^2} \int_0^\infty \left(\frac{x}{\hbar B}\right)^{2/n} \delta(x-\varepsilon) \\ &\qquad \times \frac{1}{n}(\hbar B)^{-1/n} x^{1/n-1}\, dx \\ &= \frac{V}{2\pi^2 n(\hbar B)^{3/n}} \int_0^\infty x^{3/n-1} \delta(x-\varepsilon)\, dx \\ &= \boxed{\frac{V}{2\pi^2 n(\hbar B)^{3/n}} \varepsilon^{3/n-1}}. \end{aligned}$$

Neglecting the zero-point energy, the thermal energy is given by

$$E = \int_0^\infty g(\varepsilon) \frac{\varepsilon}{e^{\beta\varepsilon} - 1}\, d\varepsilon.$$

Substituting the density of states, we obtain

$$E = \frac{V}{2\pi^2 n(\hbar B)^{3/n}} \int_0^\infty \frac{\varepsilon^{3/n}}{e^{\beta\varepsilon} - 1}\, d\varepsilon.$$

Let

$$x = \beta\varepsilon = \frac{\varepsilon}{k_B T}.$$

Then

$$\begin{aligned} E &= \frac{V(k_B T)^{1+3/n}}{2\pi^2 n(\hbar B)^{3/n}} \\ &\qquad \times \int_0^\infty \frac{x^{3/n}}{e^x - 1}\, dx. \end{aligned}$$

Since the integral is independent of temperature,

$$E \propto T^{1+3/n}.$$

Therefore, the heat capacity satisfies

$$\boxed{C = \frac{\partial E}{\partial T} \propto T^{3/n}}.$$